

Relativistic Quantum Mechanics II: Dirac Equation

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1 Dirac Equation

1.1 Dirac Equation in the Spinor and Standard Representations

In the previous note we have discussed the spinor representation of the wave function of a spin $\frac{1}{2}$ particle, and its transformation under spatial rotation, Lorentz transformation and spatial inversion. In this note, we will derive wave equations (i.e., the Dirac equation) for the spinor wave functions. The wave function of a spin $\frac{1}{2}$ particle is described by a three-dimensional spinor in its rest frame. This 3-spinor could have two different “four-dimensional origins”, ξ^α or $\eta_{\dot{\alpha}}$. Both quantities are needed for a relativistic invariant theory.

For a free particle, the only operator that could enter the wave equation is the 4-momentum operator:*

$$\hat{p}^\mu = i\partial^\mu = i\left(\frac{\partial}{\partial t}, -\nabla\right) \quad (1.1)$$

In the spinor representation, the 4-momentum operator is equivalent to a spinor operator $\hat{p}^{\alpha\dot{\beta}}$ via

$$\hat{p}^{\alpha\dot{\beta}} = \hat{p}_0 + \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} = \begin{pmatrix} \hat{p}_0 + \hat{p}_z & \hat{p}_x - i\hat{p}_y \\ \hat{p}_x + i\hat{p}_y & \hat{p}_0 - \hat{p}_z \end{pmatrix} \quad (1.2)$$

where $\boldsymbol{\sigma}$ are the Pauli matrices, $\hat{\mathbf{p}} = -i\nabla$ is the 3-momentum operator defined as in non-relativistic quantum mechanics.

The wave equation is a linear differential equation involving $\hat{p}^{\alpha\dot{\beta}}$, ξ^α , and $\eta_{\dot{\alpha}}$. The requirement of relativistic invariance leads to the following equation:

$$\boxed{\begin{aligned} \hat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} &= m\xi^\alpha \\ \hat{p}_{\dot{\beta}\alpha}\xi^\alpha &= m\eta_{\dot{\beta}} \end{aligned}} \quad (1.3)$$

where m is a dimensional constant. Substituting the second equation into the first one, and note that

$$\hat{p}^{\alpha\dot{\beta}}\hat{p}_{\dot{\beta}\gamma} = \delta_\gamma^\alpha \hat{p}^\mu \hat{p}_\mu \quad (1.4)$$

*In this note we will use relativistic units, where $c = 1$, $\hbar = 1$. In SI unit the 4-momentum operator is $\hat{p}^\mu = i\hbar\partial^\mu = i\hbar\left(\frac{1}{c}\frac{\partial}{\partial t}, -\nabla\right)$.

we have

$$\boxed{(\hat{p}^2 - m^2)\xi^\alpha = 0} \quad (1.5)$$

This second-order equation also holds for $\eta_{\dot{\alpha}}$, $(\hat{p}^2 - m^2)\eta_{\dot{\alpha}} = 0$. Clearly, m is the rest mass of the particle. Eq. (1.3) is the **Dirac equation** for spin $\frac{1}{2}$ particles.

In the spinor representation, it's apparent that Dirac equation is invariant under the transformations in the proper Lorentz group. It's also easy to show that Dirac equation is invariant under spatial inversion, where $\hat{p}^{\alpha\dot{\beta}}$, ξ^α , and $\eta_{\dot{\beta}}$ are transformed as:

$$P: \quad \hat{p}^{\alpha\dot{\beta}} \rightarrow \hat{p}_{\dot{\alpha}\beta}, \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\beta}} \rightarrow \xi^\beta \quad (1.6)$$

Note that

$$\hat{p}_{\dot{\alpha}\beta} = \hat{p}_0 - \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \quad (1.7)$$

which is easily seen from the transformation rule of the 4-spinors and 4-vectors under inversion, we can then write Dirac equation in a matrix form:

$$\boxed{\begin{aligned} (\hat{p}_0 + \boldsymbol{\sigma} \cdot \hat{\mathbf{p}})\eta &= m\xi \\ (\hat{p}_0 - \boldsymbol{\sigma} \cdot \hat{\mathbf{p}})\xi &= m\eta \end{aligned}} \quad (1.8)$$

where ξ and η are two-row column matrices whose components corresponds to ξ^α and $\eta_{\dot{\alpha}}$, respectively. The multiplication of $\boldsymbol{\sigma}$ with ξ^α and $\eta_{\dot{\alpha}}$ follows the matrix multiplication rule.

To further simplify the expression, we define the four-component bispinor ψ as

$$\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix} \quad (1.9)$$

then Eq. (1.8) can be written as

$$\begin{pmatrix} 0 & \hat{p}_0 + \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \\ \hat{p}_0 - \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = m \begin{pmatrix} \xi \\ \eta \end{pmatrix} \quad (1.10)$$

Note that all “elements” in the matrix of this expression are 2×2 blocks. Introducing the **Dirac γ matrices**:

$$\boxed{\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}} \quad (1.11)$$

γ^0 is **Hermitian** and $\boldsymbol{\gamma}$ is **anti-Hermitian**. The matrix appeared in Eq. (1.10) can be written as

$$\gamma \hat{\mathbf{p}} \equiv \gamma^\mu \hat{p}_\mu = \gamma^0 \hat{p}_0 - \boldsymbol{\gamma} \cdot \hat{\mathbf{p}} \quad (1.12)$$

Then the Dirac equation can be expressed in a very compact form

$$\boxed{(\gamma \hat{\mathbf{p}} - m)\psi = 0} \quad (1.13)$$

Note that the multiplication of γ matrices with ψ follows the matrix multiplication rule, where as the multiplication of γ^μ and $\hat{p}_\mu$ follows the multiplication rule between matrix and scalars.

Eq. (1.13) together with the γ matrices defined in Eq. (1.11) give the **Dirac equation in spinor representation**.

In general, ψ can be expressed in terms of any linear combinations of ξ^1 , ξ^2 , η_1 , and η_2 . **But we will restrict ourselves on unitary transformations of the four components, because such transformations leaves the bilinear forms constructed by ψ and ψ^* unchanged.** If ψ is expressed by different independent component, we can still have the Dirac equation being written as in Eq. (1.13), but the γ matrices will no longer have the form given in Eq. (1.11). In order to make Eq. (1.13) valid, the γ matrices must be chosen so that $\hat{p}^2 = m^2$ (Eq. (1.5)). We multiply $\gamma\hat{p}$ from the left in Eq. (1.13), then

$$(\gamma^\nu \hat{p}_\nu)(\gamma^\mu \hat{p}_\mu)\psi = m(\gamma^\nu \hat{p}_\nu)\psi = m^2\psi \quad (1.14)$$

All operators $\hat{p}_\mu$ mutually commute, $\hat{p}_\mu$ also commute with γ^μ , whereas γ^ν and γ^μ don't commute, we therefore have

$$\frac{1}{2}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) \hat{p}_\mu \hat{p}_\nu \psi = m^2\psi \quad (1.15)$$

To satisfy the requirement $\hat{p}^2 = m^2$, we must have

$$\boxed{\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}} \quad (1.16)$$

Thus **all pairs of different γ^μ anti-commute**, and their squares are

$$\boxed{(\gamma^0)^2 = 1, \quad (\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -1} \quad (1.17)$$

The ± 1 on the right side of these equations should be viewed as ± 1 multiplied by the 4×4 identity matrix. Other properties of the γ matrices derived from Eq. (1.16) will be discussed in the next section.

When ψ is transformed by a unitary transformation $\phi' = U\psi$, to make the form of Eq. (1.13) invariant, γ matrices will be transformed as $\gamma' = U\gamma U^\dagger$. Thus γ'^0 will still be Hermitian and γ' will still be anti-Hermitian. Therefore, the γ matrices in any representation have the property:

$$\boxed{\gamma^{0\dagger} = \gamma^0, \quad \gamma^\dagger = -\gamma} \quad (1.18)$$

One important unitary transformation is

$$\boxed{\phi = \frac{1}{\sqrt{2}}(\xi + \eta), \quad \chi = \frac{1}{\sqrt{2}}(\xi - \eta)} \quad (1.19)$$

In this case the γ matrices are

$$\boxed{\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 0 & \sigma \\ -\sigma & 0 \end{pmatrix}} \quad (1.20)$$

This is the **standard representation of Dirac equation**. Its advantage over the spinor representation is evident at the non-relativistic limit. Taking $\mathbf{p} \rightarrow 0$ limit in Eq. (1.8), we have $\xi = \eta$. In spinor representation, ψ will have four non-zero components but only two are independent, while in the standard representation, $\chi = 0$, thus ϕ only have two non-zero components. It's thus very clear that the two components left correspond to the 3-spinor wave function of the particle in non-relativistic theory.

1.2 Particle Current Density and Hamiltonian

Next, let's consider the wave equation for ψ^* . Taking the Hermitian adjoint of Eq. (1.13), we have

$$\psi^*(\gamma^{0\dagger}\hat{p}_0^* - \gamma^\dagger \cdot \hat{\mathbf{p}}^* - m) = 0 \quad (1.21)$$

Note that ψ^* appears on the left side of a matrix and thus should be viewed as a row matrix, and $\hat{p}_\mu$ should be viewed as acting to the left. Since $\hat{p}_\mu = -i\partial_\mu$, $\hat{p}_\mu^* = -\hat{p}_\mu$, thus

$$\psi^*(-\gamma^0\hat{p}_0 - \gamma \cdot \hat{\mathbf{p}} - m) = 0 \quad (1.22)$$

Multiply this equation by γ^0 from the right, and use the fact that $\gamma\gamma^0 = -\gamma^0\gamma$, we then have

$$\boxed{\bar{\psi}(\gamma\hat{\mathbf{p}} + m) = 0} \quad (1.23)$$

where

$$\boxed{\bar{\psi} = \psi^*\gamma^0} \quad (1.24)$$

is the **Dirac conjugate** of ψ . In the bispinor representation, multiplication with γ^0 interchanges ξ and η , thus $\bar{\psi} = (\eta^*, \xi^*)$. The first spinor in $\bar{\psi}$ has undotted indices and the second one has dotted indices, same as ψ . Thus $\bar{\psi}$ is a more natural “partner” of ψ than ψ^* .

Multiplying Eq. (1.13) by $\bar{\psi}$ from left and multiplying Eq. (1.23) by ψ from right, then adding these two equations together, we have

$$\bar{\psi}\gamma^\mu(\hat{p}_\mu\psi) + (\hat{p}_\mu\bar{\psi})\gamma^\mu\psi = i\partial_\mu(\bar{\psi}\gamma^\mu\psi) = 0 \quad (1.25)$$

Note that in Eq. (1.23) $\hat{p}_\mu$ act to the left. This equation has the form of an **equation of continuity**, $\partial_\mu j^\mu = 0$, which leads to the expression of the **particle current density 4-vector**,

$$\boxed{j^\mu = \bar{\psi}\gamma^\mu\psi = (\psi^*\psi, \psi^*\gamma^0\gamma\psi)} \quad (1.26)$$

The time component is the probability density of finding the particle at a specific location, which is positive definite.

Eq. (1.13) only contains first-order time derivative, thus we could rewrite it in the form of

$$i\frac{\partial}{\partial t}\psi = \hat{H}\psi \quad (1.27)$$

where $\hat{H}$ is the **Hamiltonian operator for the relativistic spin $\frac{1}{2}$ particle**,

$$\boxed{\hat{H} = \boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + \beta m} \quad (1.28)$$

with

$$\boxed{\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma}, \quad \beta = \gamma^0} \quad (1.29)$$

In spinor representation,

$$\boldsymbol{\alpha} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & -\boldsymbol{\sigma} \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{H} = \begin{pmatrix} \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} & m \\ m & -\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \end{pmatrix} \quad (1.30)$$

whereas in the standard representation,

$$\alpha = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{H} = \begin{pmatrix} m & \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \\ \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} & -m \end{pmatrix} \quad (1.31)$$

In both representations, the **spin operator** is given by

$$\hat{\mathbf{s}} = \frac{1}{2} \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix} \quad (1.32)$$

We calculate the commutator $[\hat{H}, \hat{s}_z]$ as an example. Using the commutation rule of the Pauli matrices

$$\sigma_i \sigma_k + \sigma_k \sigma_i = 2\delta_{ik} \quad (1.33)$$

$$\sigma_i \sigma_k = ie_{ikl} \sigma_l + \delta_{ik} \quad (1.34)$$

in spinor representation,

$$[\hat{H}, \hat{s}_z] = \begin{pmatrix} -i\sigma_y \hat{p}_x + i\sigma_x \hat{p}_y & 0 \\ 0 & i\sigma_y \hat{p}_x - i\sigma_x \hat{p}_y \end{pmatrix} \quad (1.35)$$

in standard representation,

$$[\hat{H}, \hat{s}_z] = \begin{pmatrix} 0 & -i\sigma_y \hat{p}_x + i\sigma_x \hat{p}_y \\ -i\sigma_y \hat{p}_x + i\sigma_x \hat{p}_y & 0 \end{pmatrix} \quad (1.36)$$

In general,

$$\boxed{[\hat{H}, \hat{\mathbf{s}}] \neq 0} \quad (1.37)$$

In relativistic theory, the spin operator does not commute with the Hamiltonian operator, thus the spin angular momentum and its projection is not conserved.

1.3 Dirac Matrices

In this section we will summarize some useful formulae and rules of the algebra of the Dirac matrices. These rules are derived from the general relation Eq. (1.16) and are thus not restricted to any specific forms in any particular representation.

The “**scalar product**” of the γ matrices with themselves is $g_{\mu\nu} \gamma^\mu \gamma^\nu = 4$. If we formally define the “covariant components” $\gamma_\mu \equiv g_{\mu\nu} \gamma^\nu = (\gamma^0, -\boldsymbol{\gamma})$, then

$$\gamma_\mu \gamma^\mu = 4 \quad (1.38)$$

The number “4” should be viewed as 4 times a 4×4 identity matrix. If γ_μ and γ^μ are separated by other γ matrices, the results can be calculated by applying the commutation rule Eq. (1.16). For example,

$$\begin{aligned} \gamma_\mu \gamma^\nu \gamma^\mu &= -2\gamma^\nu \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\mu &= 4g^{\lambda\nu} \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\mu &= -2\gamma^\rho \gamma^\nu \gamma^\lambda \end{aligned} \quad (1.39)$$

In practice, γ^μ often appears in the equation as a “scalar product” with a 4-vector, $\gamma a \equiv \gamma^\mu a_\mu$, like the $\gamma \hat{p}$ term in the Dirac equation. Multiply a_μ and b_ν with Eq. (1.16), we have

$$(a\gamma)(b\gamma) + (b\gamma)(a\gamma) = 2(ab) \quad (1.40)$$

Thus

$$(a\gamma)(a\gamma) = a^2 \quad (1.41)$$

Similarly, Eq. (1.39) becomes

$$\begin{aligned} \gamma_\mu(a\gamma)\gamma^\mu &= -2(a\gamma) \\ \gamma_\mu(a\gamma)(b\gamma)\gamma^\mu &= 4(ab) \\ \gamma_\mu(a\gamma)(b\gamma)(c\gamma)\gamma^\mu &= -2(c\gamma)(b\gamma)(a\gamma) \end{aligned} \quad (1.42)$$

Another frequent operation is to take the **trace of the product of a series of γ matrices**. The quantity

$$T^{\mu_1\mu_2\cdots\mu_n} = \frac{1}{4}\text{Tr}(\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}) \quad (1.43)$$

is a 4-tensor of rank n . Obviously, $T^{\mu_1\mu_2\cdots\mu_n}$ is symmetrical with respect to cyclic permutation of its indices. Since the γ matrices have the same form in any frame of reference, the components in tensor T are also independent of the frame. Only the metric tensor $g^{\mu\nu}$ has this property. Thus, $T^{\mu_1\mu_2\cdots\mu_n}$ **can be expressed entirely in terms of the metric tensor**. Furthermore, since the metric tensor is of rank 2, **only T tensors of even rank can be constructed**. This is evident because

$$\text{Tr} \gamma^\mu = 0 \quad (1.44)$$

as the trace is invariant under similarity transformations. For the case of rank 2, we take the trace for both sides of Eq. (1.16), then

$$T^{\mu\nu} = g^{\mu\nu} \quad (1.45)$$

The T tensors of higher rank can be calculated by “pulling” the first γ matrix towards the end using the commutation rule Eq. (1.16). For example,

$$\begin{aligned} T^{\lambda\mu\nu\rho} &= \frac{1}{4}\text{Tr}(\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho) \\ &= \frac{1}{4}\text{Tr}[(2g^{\lambda\mu} - \gamma^\mu\gamma^\lambda)\gamma^\nu\gamma^\rho] \\ &= 2g^{\lambda\mu}g^{\nu\rho} - \frac{1}{4}\text{Tr}(\gamma^\mu\gamma^\lambda\gamma^\nu\gamma^\rho) \\ &= \dots \\ &= 2g^{\lambda\mu}g^{\nu\rho} - 2g^{\lambda\nu}g^{\mu\rho} + 2g^{\mu\nu}g^{\lambda\rho} - T^{\mu\nu\rho\lambda} \end{aligned} \quad (1.46)$$

Using that $T^{\mu\nu\rho\lambda} = T^{\lambda\mu\nu\rho}$, we have

$$T^{\lambda\mu\nu\rho} = g^{\lambda\mu}g^{\nu\rho} - g^{\lambda\nu}g^{\mu\rho} + g^{\mu\nu}g^{\lambda\rho} \quad (1.47)$$

Tensors of higher rank can be calculated in the same manner. The formulae of the “scalar product” $(a\gamma)$ are

$$\frac{1}{4}\text{Tr}[(a\gamma)(b\gamma)] = (ab) \quad (1.48)$$

$$\frac{1}{4}\text{Tr}[(a\gamma)(b\gamma)(c\gamma)(d\gamma)] = (ab)(cd) - (ac)(bd) + (bc)(ad) \quad (1.49)$$

Sometimes we calculate the traces for γ^0 and γ separately. The formulae involving γ are

$$\frac{1}{4}\text{Tr}[(\mathbf{a} \cdot \gamma)(\mathbf{b} \cdot \gamma)] = -\mathbf{a} \cdot \mathbf{b} \quad (1.50)$$

$$\frac{1}{4}\text{Tr}[(\mathbf{a} \cdot \gamma)(\mathbf{b} \cdot \gamma)(\mathbf{c} \cdot \gamma)(\mathbf{d} \cdot \gamma)] = (\mathbf{a} \cdot \mathbf{b})(\mathbf{c} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) + (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \cdot \mathbf{d}) \quad (1.51)$$

The product $\gamma^0\gamma^1\gamma^2\gamma^3$ is of particular importance. A special notation, γ^5 is often used,[†]

$$\boxed{\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3} \quad (1.52)$$

In spinor representation,

$$\gamma^5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad (1.53)$$

In standard representation,

$$\gamma^5 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \quad (1.54)$$

We summarize the properties of γ^5 as follows. All of them are easy to prove.

$$\gamma^5\gamma^\mu + \gamma^\mu\gamma^5 = 0, \quad (\gamma^5)^2 = 1 \quad (1.55)$$

γ^5 anti-commutes with all γ^μ matrices.

$$\gamma^{5\dagger} = \gamma^5 \quad (1.56)$$

γ^5 is Hermitian.

$$\text{Tr} \gamma^5 = 0, \quad \text{Tr} (\gamma^5\gamma^\mu\gamma^\nu) = 0, \quad \frac{1}{4}\text{Tr} (\gamma^5\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho) = ie^{\lambda\mu\nu\rho} \quad (1.57)$$

2 Solutions for Free Particles

2.1 Plane Waves

The state of a free particle with definite momentum p is described by a plane wave.

For spin $\frac{1}{2}$ particles, the plane wave should have the form

$$\psi_p = \frac{1}{N}u_p e^{-ipx} \quad (2.1)$$

[†]In some literature, γ^5 is defined as $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$.

where $px \equiv p^\mu x_\mu = \varepsilon t - \mathbf{p} \cdot \mathbf{r}$ and ε is the particle's energy, $1/N$ is the normalization factor, and the wave amplitude u_p is a suitably normalized bispinor. Substituting this solution in the second-order equation for ψ ,

$$(\hat{p}^2 - m^2)\psi = 0 \quad (2.2)$$

to get a nontrivial solution, we must have $p^2 = m^2$. The energy and the momentum thus must satisfy

$$\varepsilon^2 = \mathbf{p}^2 + m^2, \quad \varepsilon = \pm \sqrt{\mathbf{p}^2 + m^2} \quad (2.3)$$

Thus we can have two possible energies for a given momentum, one positive and one negative. In classical relativistic theory, the negative energy is unphysical and could just be omitted. However, **in relativistic quantum mechanisc, the negative energy cannot be neglected, otherwise the solutions do not form a complete basis set.** We shall choose $\varepsilon = \sqrt{\mathbf{p}^2 + m^2}$ and denote the negative energy as $-\varepsilon$. The plane wave solution with negative energies may be denoted as

$$\psi_{p'} = \frac{1}{N} u_{p'} e^{-ip'x} \quad (2.4)$$

where $p'^\mu = (-\varepsilon, \mathbf{p})$. The general solution would be the linear combination of both ψ_p and $\psi_{p'}$. However, it is more convenient to also flip the sign of $\mathbf{p}$ in the negative energy solutions, then those plane waves correspond to the 4-momentum $(-\varepsilon, -\mathbf{p}) = -p$. The negative energy solutions can then be written as

$$\psi_{-p} = \frac{1}{N} u_{-p} e^{ipx} \quad (2.5)$$

It should be keep in mind that **the 4-momentum p appeared in ψ_p and ψ_{-p} is the same quantity**, $p^\mu = (\varepsilon, \mathbf{p})$. Moreover, ψ_{-p} does not equal to $\psi_{p'}$ because the space components of the 4-momentum they correspond to differs by sign. However, both ψ_{-p} and $\psi_{p'}$ form complete sets of plane waves with negative energy. The general solution of Dirac equation can instead be expressed as the linear combination of ψ_p and ψ_{-p} . **ψ_p is often referred to as the particle solution whereas ψ_{-p} is often referred to as the antiparticle solution.**

Substituting the expressions of ψ_p and ψ_{-p} into the Dirac equation, Eq. (1.13), we obtain the algebraic equations for u_p and u_{-p} ,

$$\boxed{\begin{aligned} (\gamma p - m)u_p &= (\varepsilon\gamma^0 - \mathbf{p} \cdot \boldsymbol{\gamma} - m)u_p = 0 \\ (\gamma p + m)u_{-p} &= (\varepsilon\gamma^0 - \mathbf{p} \cdot \boldsymbol{\gamma} + m)u_{-p} = 0 \end{aligned}} \quad (2.6)$$

u_p and u_{-p} satisfy different algebraic equations. We choose the **normalization conditions** for u_p and u_{-p} such that

$$\boxed{\begin{aligned} \bar{u}_p u_p &= 2m \\ \bar{u}_{-p} u_{-p} &= -2m \end{aligned}} \quad (2.7)$$

Multiplying the equations in Eq. (2.6) by $\bar{u}_p$ or $\bar{u}_{-p}$ from left, we have

$$\begin{aligned} (\bar{u}_p \gamma u_p)p &= m \bar{u}_p u_p = 2m^2 = 2p^2 \\ (\bar{u}_{-p} \gamma u_{-p})p &= -m \bar{u}_{-p} u_{-p} = 2m^2 = 2p^2 \end{aligned} \quad (2.8)$$

where we have used the condition $p^2 = m^2$. Thus

$$\bar{u}_p \gamma u_p = \bar{u}_{-p} \gamma u_{-p} = 2p \quad (2.9)$$

The 4-current is given by

$$j^\mu = \bar{\psi} \gamma^\mu \psi = \bar{u}_{\pm p} \gamma^\mu u_{\pm p} = \frac{2p^\mu}{N^2} \quad (2.10)$$

We shall **normalize the wave function to one particle in the volume $V = 1$** , then

$$\boxed{j^\mu = \bar{\psi} \gamma^\mu \psi = \left(1, \frac{\mathbf{p}}{\varepsilon}\right)} \quad (2.11)$$

where $\mathbf{p}/\varepsilon$ is the velocity of the particle. Therefore, we find the normalization factor

$$\frac{1}{N} = \frac{1}{\sqrt{2\varepsilon}} \quad (2.12)$$

The plane wave solutions are thus

$$\boxed{\psi_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}, \quad \psi_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}} \quad (2.13)$$

The actual u_p and u_{-p} as solutions of Eq. (2.6) depend on the specific representation of choice. Here we focus on the standard representation. For u_p , Eq. (2.6) becomes

$$\begin{aligned} (\varepsilon - m)\phi - (\mathbf{p} \cdot \boldsymbol{\sigma})\chi &= 0 \\ (\varepsilon + m)\chi - (\mathbf{p} \cdot \boldsymbol{\sigma})\phi &= 0 \end{aligned} \quad (2.14)$$

Thus

$$\phi = \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{\varepsilon - m} \chi, \quad \chi = \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{\varepsilon + m} \phi \quad (2.15)$$

The common factor of ϕ and χ is chosen so that the normalization condition Eq. (2.7) is satisfied, then

$$\phi^* \phi - \chi^* \chi = 2m \quad (2.16)$$

$$\phi^* \phi = \varepsilon + m, \quad \chi^* \chi = \varepsilon - m \quad (2.17)$$

Therefore,

$$\boxed{u_p = \begin{pmatrix} \sqrt{\varepsilon + m} w \\ \sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w \end{pmatrix}} \quad (2.18)$$

where $\mathbf{n}_p$ is the unit vector along the direction of $\mathbf{p}$. **w is an arbitrary two-component quantity that satisfies $w^* w = 1$.** To arrive at the final expression, we have used the relation $|\mathbf{p}| = \sqrt{\varepsilon^2 - m^2}$. Following the same procedure, we have the expression for u_{-p} ,

$$\boxed{u_{-p} = \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w' \\ \sqrt{\varepsilon + m} w' \end{pmatrix}} \quad (2.19)$$

where w' is another two-component quantity. The final expressions for ψ_p and ψ_{-p} in standard representation are

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} w \\ \sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w \end{pmatrix} e^{-ipx}, \quad \psi_{-p} = \frac{1}{\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w' \\ \sqrt{\varepsilon + m} w' \end{pmatrix} e^{ipx} \quad (2.20)$$

The wave functions ψ_p and ψ_{-p} both contain an arbitrary two-component quantity. **For a given momentum, there are two independent states, corresponding to the two possible values of the spin component.** In the rest frame, $\varepsilon = m$, $\mathbf{p} = 0$ then

$$\psi_p = \begin{pmatrix} w \\ 0 \end{pmatrix} e^{-imt}, \quad \psi_{-p} = \begin{pmatrix} 0 \\ w' \end{pmatrix} e^{imt} \quad (2.21)$$

Clearly, w and w' correspond to the **3-spinor wave function of the particle in the rest frame**. This correspondence implies that w' and w must be related to each other through a certain transformation in the spinor space. Otherwise, the linear combination of ψ_p and ψ_{-p} will have four independent components.

We can choose the two independent states of the particle to correspond to the two values of the z -component of the spin vector in the rest frame, for ψ_p , we have

$$w^{(\sigma=\frac{1}{2})} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad w^{(\sigma=-\frac{1}{2})} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2.22)$$

For ψ_{-p} , w' and w are related via

$$w'^{(\sigma)} = -\sigma_y w^{(-\sigma)} \quad (2.23)$$

thus

$$w'^{(\sigma=\frac{1}{2})} = \begin{pmatrix} i \\ 0 \end{pmatrix}, \quad w'^{(\sigma=-\frac{1}{2})} = \begin{pmatrix} 0 \\ -i \end{pmatrix} \quad (2.24)$$

This choice will be rationalized in subsequent sections.

The z -component of the particle's spin vector does not have a definite value in other frames because the spin operator does not commute with the Hamiltonian. However, **the projection of the spin operator on the direction of the momentum, $\hat{\mathbf{s}} \cdot \mathbf{n}_p$, does commute with the Hamiltonian and thus the spin component along the direction of $\mathbf{p}$ can have definite values.** This quantity is called the **helicity** of the particle. Helicity states corresponds to the plane waves in which the 3-spinor is an eigenvector of the operator $\hat{\mathbf{s}} \cdot \mathbf{n}_p$,

$$(\hat{\mathbf{s}} \cdot \mathbf{n}_p) w^{(\lambda)} = \lambda w^{(\lambda)}, \quad \lambda = \pm \frac{1}{2} \quad (2.25)$$

Let θ and ϕ be the polar angle and azimuth of the vector $\mathbf{n}_p$ relative to a fixed xyz coordinate, then

$$w^{(\lambda=\frac{1}{2})} = \begin{pmatrix} e^{-\frac{1}{2}i\phi} \cos \frac{\theta}{2} \\ e^{\frac{1}{2}i\phi} \sin \frac{\theta}{2} \end{pmatrix}, \quad w^{(\lambda=-\frac{1}{2})} = \begin{pmatrix} -e^{-\frac{1}{2}i\phi} \sin \frac{\theta}{2} \\ e^{\frac{1}{2}i\phi} \cos \frac{\theta}{2} \end{pmatrix} \quad (2.26)$$

The expressions for $w'^{(\lambda)}$ can be obtained by setting θ and ϕ to the polar angle and azimuth of the vector $\mathbf{n}_{-p}$, since ψ_{-p} corresponds to a reversed 3-momentum. Substituting $\theta \rightarrow \pi - \theta$, $\phi \rightarrow \phi + \pi$,

$$w'^{(\lambda=\frac{1}{2})} = i \begin{pmatrix} -e^{-\frac{1}{2}i\phi} \sin \frac{\theta}{2} \\ e^{\frac{1}{2}i\phi} \cos \frac{\theta}{2} \end{pmatrix}, \quad w'^{(\lambda=-\frac{1}{2})} = i \begin{pmatrix} e^{-\frac{1}{2}i\phi} \cos \frac{\theta}{2} \\ e^{\frac{1}{2}i\phi} \sin \frac{\theta}{2} \end{pmatrix} \quad (2.27)$$

Thus

$$w'^{(\lambda)} = i w^{(-\lambda)} \quad (2.28)$$

Here $i = e^{i\frac{\pi}{2}}$ is just a constant phase factor, which is unimportant. This result is obvious, as reversing the direction of $\mathbf{p}$ only changes the sign of $\hat{\mathbf{s}} \cdot \mathbf{n}_p$.

2.2 Particles and Antiparticles

In quantum field theory, “particles” are the excitations of the corresponding field. In this section we derive the second quantization formulae for the field of spin $\frac{1}{2}$ particles, i.e., the **spinor field**.

Recall in the second quantization formalism of the electromagnetic field, in the transverse (Coulomb) gauge, the Maxwell equation reduced to the wave equation for the vector potential $\mathbf{A}$,

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0 \quad (2.29)$$

The general solution is given by linear combination of plane waves,

$$\mathbf{A} = \sum_{\mathbf{k}, \lambda} \left(\mathbf{a}_{\mathbf{k}\lambda} e^{i\mathbf{k} \cdot \mathbf{r} - i\omega t} + \mathbf{a}_{\mathbf{k}\lambda}^* e^{-i\mathbf{k} \cdot \mathbf{r} + i\omega t} \right) \quad (2.30)$$

This solution $\mathbf{A}$ can be viewed as the wave function of the photon, whereas the expansion coefficients $\mathbf{a}_{\mathbf{k}\lambda}$ and $\mathbf{a}_{\mathbf{k}\lambda}^*$ can be viewed as the discrete “field variables”. $\mathbf{a}_{\mathbf{k}\lambda} + \mathbf{a}_{\mathbf{k}\lambda}^*$ and $\mathbf{a}_{\mathbf{k}\lambda} - \mathbf{a}_{\mathbf{k}\lambda}^*$ corresponds to the canonical field coordinate and momentum. The quantization is achieved by replacing $\mathbf{a}_{\mathbf{k}\lambda}$ and $\mathbf{a}_{\mathbf{k}\lambda}^*$ by the annihilation and creation operators $\hat{a}_{\mathbf{k}\lambda}$ and $\hat{a}_{\mathbf{k}\lambda}^\dagger$. $\mathbf{A}$ thus becomes a field operator $\hat{\mathbf{A}}$. The operators of other field observables like $\mathbf{E}$ and $\mathbf{B}$ can be derived by replacing $\mathbf{A}$ with $\hat{\mathbf{A}}$ in the corresponding expressions. The quantization of a spinor field follows a similar procedure.

The wave equation of a spinor field is the Dirac equation. Its general solution is the linear combination of ψ_p and ψ_{-p} . Specifying the particle’s polarization σ (spin component in the rest frame) for each plane wave, the spinor field is given by

$$\psi = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(a_{\mathbf{p}\sigma}^{(+)} u_{\mathbf{p}\sigma} e^{-ipx} + a_{\mathbf{p}\sigma}^{(-)} u_{-\mathbf{p}, -\sigma} e^{ipx} \right) \quad (2.31)$$

The summation is over the 3-momentum $\mathbf{p}$ and $\sigma = \pm \frac{1}{2}$. $a_{\mathbf{p}\sigma}^{(+)}$ and $a_{\mathbf{p}\sigma}^{(-)}$ are the expansion coefficients for the plane waves with positive and negative energies, respectively. Unlike the case for electromagnetic field, $a_{\mathbf{p}\sigma}^{(+)}$ and $a_{\mathbf{p}\sigma}^{(-)}$ **are not the complex conjugate of each other but are independent field variables**.

Note that the negative energy solution for a given polarization σ in rest frame is $\psi_{-p,-\sigma}$ instead of $\psi_{-p,\sigma}$, because $\psi_{-p,-\sigma}$ can be transformed to $\bar{\psi}_{p\sigma}$ via a unitary transformation. The later clearly corresponds to a state with polarization σ . In standard representation, the transformation matrix is

$$U_C \equiv \gamma^2 \gamma^0 = -\alpha_y \quad (2.32)$$

Using the expressions of $u_{p\sigma}$, $u_{-p,-\sigma}$ given in the previous section, we can show that

$$\begin{aligned} U_C u_{-p,-\sigma} &= \begin{pmatrix} 0 & -\sigma_y \\ -\sigma_y & 0 \end{pmatrix} \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w'^{(-\sigma)} \\ \sqrt{\varepsilon + m} w'^{(-\sigma)} \end{pmatrix} \\ &= \begin{pmatrix} -\sqrt{\varepsilon + m} \sigma_y w'^{(-\sigma)} \\ -\sqrt{\varepsilon - m} \sigma_y (\mathbf{n}_p \cdot \boldsymbol{\sigma}) w'^{(-\sigma)} \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{\varepsilon + m} w^{*(\sigma)} \\ -\sqrt{\varepsilon - m} (\mathbf{n}_p \cdot \boldsymbol{\sigma}^*) w^{*(\sigma)} \end{pmatrix} \\ &= (\sqrt{\varepsilon + m} w^{*(\sigma)}, -\sqrt{\varepsilon - m} w^{*(\sigma)} (\mathbf{n}_p \cdot \boldsymbol{\sigma}))^T \\ &= (\bar{u}_{p\sigma})^T \end{aligned} \quad (2.33)$$

where we also used the relation between $w'^{(-\sigma)}$ and $w^{(\sigma)}$ defined in Eq. (2.23), $w^{(\sigma)} = w^{*(\sigma)}$, and

$$\sigma_y \boldsymbol{\sigma} = -\boldsymbol{\sigma}^* \sigma_y, \quad \sigma_y^2 = 1 \quad (2.34)$$

In summary,

$$U_C u_{-p,-\sigma} = (\bar{u}_{p\sigma})^T, \quad U_C (\bar{u}_{-p,-\sigma})^T = u_{p\sigma} \quad (2.35)$$

Thus

$$U_C \psi_{-p,-\sigma} = (\bar{\psi}_{p\sigma})^T, \quad U_C (\bar{\psi}_{-p,-\sigma})^T = \psi_{p\sigma} \quad (2.36)$$

In second quantization theory, the time dependence $e^{-i\varepsilon t}$ corresponds to particle annihilation while its conjugate $e^{i\varepsilon t} = (e^{-i\varepsilon t})^*$ corresponds to particle creation. Since $a_{p\sigma}^{(+)}$ and $a_{p\sigma}^{(-)}$ are independent field variables, upon second quantization, $a_{p\sigma}^{(+)}$ will be replaced by the annihilation operator $\hat{a}_{p\sigma}$ as usual, but $a_{p\sigma}^{(-)}$ must be replaced by the creation operator of a second kind of particle, $\hat{b}_{p\sigma}^\dagger$. The quantized field operator and its Dirac conjugate are thus

$$\hat{\psi} = \sum_{p,\sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{p\sigma} u_{p\sigma} e^{-ipx} + \hat{b}_{p\sigma}^\dagger u_{-p,-\sigma} e^{ipx} \right) \quad (2.37)$$

$$\hat{\bar{\psi}} = \sum_{p,\sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{p\sigma}^\dagger \bar{u}_{p\sigma} e^{ipx} + \hat{b}_{p\sigma} \bar{u}_{-p,-\sigma} e^{-ipx} \right) \quad (2.38)$$

We therefore arrived at two types of particles, corresponding to the creation and annihilation operators $\hat{a}_{p\sigma}/\hat{a}_{p\sigma}^\dagger$ and $\hat{b}_{p\sigma}/\hat{b}_{p\sigma}^\dagger$, respectively. They are called **particles** and **antiparticles**. The particles and antiparticles have the same mass, as they appear in the same field operator.

Next, we derive the commutation rules for $\hat{a}_{\mathbf{p}\sigma}$ and $\hat{a}_{\mathbf{p}\sigma}^\dagger$ as well as $\hat{b}_{\mathbf{p}\sigma}$ and $\hat{b}_{\mathbf{p}\sigma}^\dagger$. For photons, the commutation rules are obtained from the commutation rule of the canonical field coordinate and momentum, $\hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda}^\dagger$ and $\hat{a}_{\mathbf{k}\lambda} - \hat{a}_{\mathbf{k}\lambda}^\dagger$. The applicability of the concepts of canonical field coordinate and momentum is a property of the electromagnetic field, which can not be applied for spinor fields. **Instead, we derive the commutation rules based on the requirement that the total energy of the field must be positive definite.**

The average energy of the system with wave function ψ is given by

$$H = \int \psi^* \hat{H} \psi d\mathbf{r} = i \int \bar{\psi} \gamma^0 \frac{\partial \psi}{\partial t} d\mathbf{r} \quad (2.39)$$

where H denotes the Hamiltonian function and $\hat{H}$ is the Hamiltonian operator. Replacing ψ by $\hat{\psi}$ we have the **Hamiltonian operator in the second quantization formalism**,

$$\hat{H} = \sum_{\mathbf{p}, \sigma} \varepsilon \left(\hat{a}_{\mathbf{p}\sigma}^\dagger \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma} \hat{b}_{\mathbf{p}\sigma}^\dagger \right) \quad (2.40)$$

If $\hat{a}_{\mathbf{p}\sigma}/\hat{a}_{\mathbf{p}\sigma}^\dagger$ and $\hat{b}_{\mathbf{p}\sigma}/\hat{b}_{\mathbf{p}\sigma}^\dagger$ follows Fermi anti-commutation rules,

$$\boxed{\{\hat{a}_{\mathbf{p}\sigma}, \hat{a}_{\mathbf{p}\sigma}^\dagger\} = 1, \quad \{\hat{b}_{\mathbf{p}\sigma}, \hat{b}_{\mathbf{p}\sigma}^\dagger\} = 1} \quad (2.41)$$

where $\{, \}$ denotes the anti-commutator, then the Hamiltonian can be rewritten as

$$\boxed{\hat{H} = \sum_{\mathbf{p}, \sigma} \varepsilon \left(\hat{a}_{\mathbf{p}\sigma}^\dagger \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma}^\dagger \hat{b}_{\mathbf{p}\sigma} - 1 \right)} \quad (2.42)$$

Its eigenvalues are thus

$$\boxed{E = \sum_{\mathbf{p}, \sigma} \varepsilon (N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma})} \quad (2.43)$$

where $N_{\mathbf{p}\sigma}$ and $\bar{N}_{\mathbf{p}\sigma}$ are the number of particles and antiparticles, respectively. The infinite zero-point energy has been omitted. Clearly $E \geq 0$. Therefore, if $\hat{a}_{\mathbf{p}\sigma}/\hat{a}_{\mathbf{p}\sigma}^\dagger$ and $\hat{b}_{\mathbf{p}\sigma}/\hat{b}_{\mathbf{p}\sigma}^\dagger$ follows Fermi anti-commutation rules, the total energy of the field will be positive definite. Conversely, if $\hat{a}_{\mathbf{p}\sigma}/\hat{a}_{\mathbf{p}\sigma}^\dagger$ and $\hat{b}_{\mathbf{p}\sigma}/\hat{b}_{\mathbf{p}\sigma}^\dagger$ follows Bose commutation rules, $[\hat{a}_{\mathbf{p}\sigma}, \hat{a}_{\mathbf{p}\sigma}^\dagger] = 1$, $[\hat{b}_{\mathbf{p}\sigma}, \hat{b}_{\mathbf{p}\sigma}^\dagger] = 1$, we would arrive at $E = \sum_{\mathbf{p}, \sigma} \varepsilon (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma})$ which is not positive definite and have no meaning. Therefore, we conclude that **the spin $\frac{1}{2}$ particles and antiparticles are Fermions.**

The **total momentum operator of the system** in second quantization formalism is given by

$$\hat{\mathbf{P}} = \int \hat{\psi}^\dagger \hat{\mathbf{p}} \hat{\psi} d\mathbf{r} = i \int \hat{\bar{\psi}} \gamma^0 \nabla \hat{\psi} d\mathbf{r} = \sum_{\mathbf{p}, \sigma} \mathbf{p} \left(\hat{a}_{\mathbf{p}\sigma}^\dagger \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma}^\dagger \hat{b}_{\mathbf{p}\sigma} - 1 \right) \quad (2.44)$$

with eigenvalues

$$\boxed{\mathbf{P} = \sum_{\mathbf{p}, \sigma} \mathbf{p} (N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma})} \quad (2.45)$$

The 4-current density operator is

$$\hat{j}^\mu = \hat{\bar{\psi}} \gamma^\mu \hat{\psi} \quad (2.46)$$

Then the **total “charge” operator of the system** is

$$\hat{Q} = \int \hat{j}^0 d\mathbf{r} = \int \hat{\psi} \gamma^0 \hat{\psi} d\mathbf{r} = \sum_{\mathbf{p}, \sigma} \left(\hat{a}_{\mathbf{p}\sigma}^\dagger \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma}^\dagger \hat{b}_{\mathbf{p}\sigma} + 1 \right) \quad (2.47)$$

Its eigenvalues are

$$Q = \sum_{\mathbf{p}, \sigma} (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}) \quad (2.48)$$

Note that Q is not the electric charge of the field, and $\hat{j}^0$ can not be interpreted as the probability density of the spatial localization of the particle because it can be negative.

For free particles with no interaction between them, the particle numbers $N_{\mathbf{p}\sigma}$ and $\bar{N}_{\mathbf{p}\sigma}$ are conserved quantities. Thus Q is also conserved. For interacting particles, whether Q is conserved depends on the nature of the interaction. **If the operator $\hat{Q}$ commute with the Hamiltonian of the interaction $\hat{H}_{\text{int}}$, then Q is conserved. For such cases the particles and antiparticles cannot be singly created or annihilated, only the particle-antiparticle pair can be created or annihilated.** The interaction of charged particle with electromagnetic field is an example. Only the particle-antiparticle pair can be created or annihilated indicates that **the antiparticle must have the opposite electric charge to the particle. Otherwise the total electric charge will not be conserved.**

The fact that spin $\frac{1}{2}$ particles are Fermions leads to the following general conclusion. **All half-integer-spin particles are Fermions and all integer-spin particles are Bosons.** Because a spin s particle can be viewed as a composite of $2s$ spin $\frac{1}{2}$ particles. According to the exchange symmetry, a composite particle contains odd number of Fermions is still a Fermion, whereas a composite particle contains even number of Fermions is a Boson.

Finally, we comment on the connection of the concept of antiparticle and the spin-statistics relation with the requirement of relativistic invariance. The relativistic energy-momentum relation $\varepsilon^2 = \mathbf{p}^2 + m^2$ leads to the positive and negative energy solutions of the equation of motion. The negative energy solution corresponds to the antiparticles. On the other hand, the wave functions of particles with different spin is described by spinors of different rank. The relativistic invariance results in different forms of the equation of motion for particles with different spin, which in turn leads to different expressions of the Hamiltonian operator in second quantization formalism. The commutation rule and statistics of the particle is determined by the requirement that the total energy must be positive definite. Thus we may say that **both the emergence of antiparticles and the spin-statistics relation are consequences of the relativistic invariance requirement.**

3 CPT Transformations

3.1 Parity Transformation

Lastly, we discuss three important discrete symmetry operations, **charge conjugate (C)**, **parity transformation (P)**, and **time reversal (T)**, and the transformation of the spinor field under these operations.

The transformation under parity or spatial inversion is straightforward. In the spinor representation, spatial inversion interchanges the two 4-spinors ξ and η :

$$P : \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha \quad (3.1)$$

The transformation of ψ in spinor representation can be expressed using γ^0 as

$$\hat{P}\psi(t, \mathbf{r}) = i\gamma^0\psi(t, -\mathbf{r}) \quad (3.2)$$

It's easy to show that this transformation rule still holds in other representations, where ψ undergoes a unitary transformation $\psi' = U\psi$, $\gamma'^0 = U\gamma^0U^\dagger$. Thus this transformation rule is general and not limited to any particular representation.

The transformed field operator is

$$\hat{\psi}^P(t, \mathbf{r}) = i\gamma^0\hat{\psi}(t, -\mathbf{r}) \quad (3.3)$$

Inserting the explicit expression of ψ , we have

$$\hat{\psi}^P = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(i\hat{a}_{\mathbf{p}\sigma}\gamma^0 u_{\mathbf{p}\sigma} e^{-i\mathbf{p}\cdot\mathbf{r} - i\varepsilon t} + i\hat{b}_{\mathbf{p}\sigma}^\dagger \gamma^0 u_{-\mathbf{p}, -\sigma} e^{i\mathbf{p}\cdot\mathbf{r} + i\varepsilon t} \right) \quad (3.4)$$

If we substitute $\mathbf{p}$ by $-\mathbf{p}$ in the expression, the summation will be unchanged ($\sum_{\mathbf{p}} = \sum_{-\mathbf{p}}$). Therefore $\hat{\psi}^P$ can be expressed as

$$\hat{\psi}^P = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(i\hat{a}_{-\mathbf{p}\sigma}\gamma^0 u_{\mathbf{p}'\sigma} e^{i\mathbf{p}\cdot\mathbf{r} - i\varepsilon t} + i\hat{b}_{-\mathbf{p}\sigma}^\dagger \gamma^0 u_{-\mathbf{p}', -\sigma} e^{-i\mathbf{p}\cdot\mathbf{r} + i\varepsilon t} \right) \quad (3.5)$$

Here $\mathbf{p}'^\mu = (\varepsilon, -\mathbf{p})$ is the transformed 4-momentum under spatial inversion. In standard representation, we can easily show that

$$\gamma^0 u_{\mathbf{p}'\sigma} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{\varepsilon + m} w^{(\sigma)} \\ \sqrt{\varepsilon - m} (\mathbf{n}_{-\mathbf{p}} \cdot \boldsymbol{\sigma}) w^{(\sigma)} \end{pmatrix} = \begin{pmatrix} \sqrt{\varepsilon + m} w^{(\sigma)} \\ \sqrt{\varepsilon - m} (\mathbf{n}_{\mathbf{p}} \cdot \boldsymbol{\sigma}) w^{(\sigma)} \end{pmatrix} = u_{\mathbf{p}\sigma} \quad (3.6)$$

This relation also holds in other representations. Similarly

$$\gamma^0 u_{-\mathbf{p}', -\sigma} = -u_{-\mathbf{p}, -\sigma} \quad (3.7)$$

Thus,

$$\hat{\psi}^P = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(i\hat{a}_{-\mathbf{p}\sigma} u_{\mathbf{p}\sigma} e^{i\mathbf{p}\cdot\mathbf{r} - i\varepsilon t} - i\hat{b}_{-\mathbf{p}\sigma}^\dagger u_{-\mathbf{p}, -\sigma} e^{-i\mathbf{p}\cdot\mathbf{r} + i\varepsilon t} \right) \quad (3.8)$$

Comparing to the field operator before transformation,

$$\hat{\psi} = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\sigma} u_{\mathbf{p}\sigma} e^{i\mathbf{p}\cdot\mathbf{r} - i\varepsilon t} + \hat{b}_{\mathbf{p}\sigma}^\dagger u_{-\mathbf{p}, -\sigma} e^{-i\mathbf{p}\cdot\mathbf{r} + i\varepsilon t} \right) \quad (3.9)$$

we find the effect of parity transformation (spatial inversion) on the field operator,

$$P : \quad \hat{a}_{\mathbf{p}\sigma} \rightarrow i\hat{a}_{-\mathbf{p}\sigma}, \quad \hat{b}_{\mathbf{p}\sigma} \rightarrow i\hat{b}_{-\mathbf{p}\sigma} \quad (3.10)$$

Parity transformation replaces the particle (antiparticle) with momentum $\mathbf{p}$ and polarization σ by a particle (antiparticle) with momentum $-\mathbf{p}$ and the same polarization σ . This result is easy to understand, as spatial inversion changes the sign of the polar vector $\mathbf{p}$ but leaves the axial vector $\mathbf{s}$ unchanged.

3.2 Charge Conjugation

In the field operator $\hat{\psi}$ and $\hat{\bar{\psi}}$, the coefficient $\psi_{p\sigma} = u_{p\sigma}e^{-ipx}$ appears with the operator $\hat{a}_{p\sigma}$ is the wave function of the particle with momentum $\mathbf{p}$ and polarization σ ; similarly, the coefficient $\bar{\psi}_{-p,-\sigma} = \bar{u}_{-p,-\sigma}e^{-ipx}$ appears with the operator $\hat{b}_{p\sigma}$ should be the wave function of the antiparticle with momentum $\mathbf{p}$ and polarization σ . $\bar{\psi}_{-p,-\sigma}$ and $\psi_{p\sigma}$ satisfy different set of equations and their transformation properties are different. However, in the previous section, we have shown that they are related via a unitary transformation,

$$U_C\psi_{-p,-\sigma} = (\bar{\psi}_{p\sigma})^T, \quad U_C(\bar{\psi}_{-p,-\sigma})^T = \psi_{p\sigma} \quad (3.11)$$

In standard and spinor representations, $U_C = \gamma^2\gamma^0 = -\alpha_y$. In other representation, U_C only need to satisfy the requirement that the transformed wave function $U_C(\bar{\psi}_{-p,-\sigma})^T$ obeys the same equation as $\psi_{p\sigma}$. The Dirac equation for $\bar{\psi}$ is

$$(\gamma^T p + m)\bar{\psi}^T = 0 \quad (3.12)$$

Multiplying U_C from the left, we have

$$U_C\gamma^T p\bar{\psi}^T + mU_C\bar{\psi}^T = 0 \quad (3.13)$$

whereas $U_C\bar{\psi}^T$ should satisfy the equation

$$(\gamma p - m)(U_C\bar{\psi}^T) = 0 \quad (3.14)$$

Comparing these two expressions we find the U_C must satisfy

$$\boxed{U_C(\gamma^\mu)^T = -\gamma^\mu U_C} \quad (3.15)$$

Using the explicit expressions of the γ matrices in spinor and standard representations, we find that this condition is only satisfied by the matrix $\eta_C\gamma^2\gamma^0$, where η_C is a phase factor, $|\eta_C| = 1$. We shall simply choose $\eta_C = 1$.[‡]

The same transformation discussed above can be applied to non-plane wave functions. **The charge conjugation is an operation which when applied to ψ returns $U_C\bar{\psi}$:**

$$\boxed{\hat{C}\psi(t, \mathbf{r}) = U_C\bar{\psi}(t, \mathbf{r})} \quad (3.16)$$

From hereon, when $\bar{\psi}$ appears on the right side of a matrix, it should be viewed as a column matrix, and the transpose sign will be omitted for simplicity. If ψ is subject to a unitary transformation, $\psi' \rightarrow U\psi$, then the transformation matrix in the new representation is

$$U'_C = UU_CU^T \quad (3.17)$$

whereas the matrix $\gamma^2\gamma^0$ in the new representation is $\gamma'^2\gamma'^0 = U\gamma^2\gamma^0U^\dagger$. Therefore, **Eq. (2.32) does not hold in other representations unless the transformation matrix U is real.**

[‡] $\eta_C = 1$ is used in Landau's QED book. Other literature may use different η_C .

In spinor representation, we may write the charge conjugation operation as

$$\hat{C}\psi(t, \mathbf{r}) = \gamma^2 \psi^*(t, \mathbf{r}) \quad (3.18)$$

Then we find the transformations of 4-spinors ξ and η under charge conjugation,

$$C : \quad \xi^\alpha \rightarrow -i\eta^{\dot{\alpha}*}, \quad \eta_{\dot{\alpha}} \rightarrow -i\xi_\alpha^* \quad (3.19)$$

$$C : \quad \xi_\alpha \rightarrow -i\eta_{\dot{\alpha}}^*, \quad \eta^{\dot{\alpha}} \rightarrow -i\xi^{\alpha*} \quad (3.20)$$

The transformed field operator is

$$\hat{\psi}^C(t, \mathbf{r}) = U_C \hat{\bar{\psi}}(t, \mathbf{r}) \quad (3.21)$$

Substituting the explicit expression of $\hat{\bar{\psi}}(t, \mathbf{r})$, and using the transformation rules of $\bar{u}_{\pm p, \pm \sigma}$ under U_C , we have

$$\hat{\psi}^C = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\sigma}^\dagger u_{-p, -\sigma} e^{ipx} + \hat{b}_{\mathbf{p}\sigma} u_{p\sigma} e^{-ipx} \right) \quad (3.22)$$

Comparing to the field operator before transformation,

$$\hat{\psi} = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\sigma} u_{p\sigma} e^{-ipx} + \hat{b}_{\mathbf{p}\sigma}^\dagger u_{-p, -\sigma} e^{ipx} \right) \quad (3.23)$$

we immediately find the effect of charge conjugation on the field operator,

$$C : \quad \hat{a}_{\mathbf{p}\sigma} \rightarrow \hat{b}_{\mathbf{p}\sigma}, \quad \hat{b}_{\mathbf{p}\sigma} \rightarrow \hat{a}_{\mathbf{p}\sigma} \quad (3.24)$$

Charge conjugation interchanges particles and antiparticles. This transformation has no non-relativistic analogue.

3.3 Time Reversal

In quantum mechanics, if a certain state is described by $\psi(t, \mathbf{r})$, then the time-reversed state is described by $\psi^*(-t, \mathbf{r})$. The complex conjugation is essential to restore the “correct” time dependence after changing the sign of t , so that $\psi^*(-t, \mathbf{r})$ satisfies the same equation as $\psi(t, \mathbf{r})$. For spin $\frac{1}{2}$ particles, $\psi^*(-t, \mathbf{r})$ obeys different equations and is transformed differently from $\psi(t, \mathbf{r})$. Similar to the charge conjugation transformation, we seek a suitable unitary transformation U_T so that $U_T \bar{\psi}(-t, \mathbf{r})$ satisfies the same equation as ψ .

The time reversal transformation is given by

$$\hat{T}\psi(t, \mathbf{r}) = U_T \bar{\psi}(-t, \mathbf{r}) \quad (3.25)$$

Following the same procedure in the derivation of U_C , the Dirac equation for $\bar{\psi}(t, \mathbf{r})$ is

$$\left(i(\gamma^0)^T \frac{\partial}{\partial t} + i\boldsymbol{\gamma}^T \cdot \nabla + m \right) \bar{\psi}(t, \mathbf{r}) = 0 \quad (3.26)$$

Changing $t \rightarrow -t$ in the equation, and multiply U_T from left, we have

$$\left(iU_T(\gamma^0)^T \frac{\partial}{\partial t} - iU_T \boldsymbol{\gamma}^T \cdot \nabla \right) \bar{\psi}(-t, \mathbf{r}) - mU_T \bar{\psi}(-t, \mathbf{r}) = 0 \quad (3.27)$$

On the other hand, $U_T \bar{\psi}(-t, \mathbf{r})$ should obey the equation

$$\left(i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \cdot \nabla - m \right) (U_T \bar{\psi}(-t, \mathbf{r})) = 0 \quad (3.28)$$

We thus find the requirement of U_T ,

$$\boxed{U_T(\gamma^0)^T = \gamma^0 U_T, \quad U_T \boldsymbol{\gamma}^T = -\boldsymbol{\gamma} U_T} \quad (3.29)$$

In spinor and standard representations, using the explicit expressions of the γ matrices, this condition is satisfied by the matrix

$$\boxed{U_T = i\gamma^3 \gamma^1 \gamma^0} \quad (3.30)$$

Here i is a phase factor. The choice of this phase factor is determined by the choice of the phase factor η_C in U_C . The transformations of 4-spinors ξ and η under time reversal are thus,

$$\boxed{T: \quad \xi^\alpha \rightarrow -i\xi_\alpha^*, \quad \eta_{\dot{\alpha}} \rightarrow i\eta^{\dot{\alpha}*}} \quad (3.31)$$

$$\boxed{T: \quad \xi_\alpha \rightarrow i\xi^{\alpha*}, \quad \eta^{\dot{\alpha}} \rightarrow -i\eta_{\dot{\alpha}}^*} \quad (3.32)$$

The transformed field operator is

$$\boxed{\hat{\psi}^T(t, \mathbf{r}) = U_T \hat{\bar{\psi}}(-t, \mathbf{r})} \quad (3.33)$$

Substituting the explicit expression of $\hat{\bar{\psi}}$, we have

$$\hat{\psi}^T = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\sigma}^\dagger U_T \bar{u}_{\mathbf{p}\sigma} e^{-i\mathbf{p} \cdot \mathbf{r} - i\varepsilon t} + \hat{b}_{\mathbf{p}\sigma} U_T \bar{u}_{-\mathbf{p}, -\sigma} e^{i\mathbf{p} \cdot \mathbf{r} + i\varepsilon t} \right) \quad (3.34)$$

Changing $\mathbf{p} \rightarrow -\mathbf{p}$ and $\sigma \rightarrow -\sigma$, then

$$\hat{\psi}^T = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{-\mathbf{p}, -\sigma}^\dagger U_T \bar{u}_{\mathbf{p}', -\sigma} e^{i\mathbf{p} \cdot \mathbf{r} - i\varepsilon t} + \hat{b}_{-\mathbf{p}, -\sigma} U_T \bar{u}_{-\mathbf{p}'\sigma} e^{-i\mathbf{p} \cdot \mathbf{r} + i\varepsilon t} \right) \quad (3.35)$$

where $p'^\mu = (\varepsilon, -\mathbf{p})$. Next, we calculate $U_T \bar{u}_{\pm \mathbf{p}', \mp \sigma}$. In standard representation,

$$U_T = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix} \quad (3.36)$$

$$(\bar{u}_{-\mathbf{p}'\sigma})^T = \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_{-\mathbf{p}} \cdot \boldsymbol{\sigma}^*) w'^*(\sigma) \\ -\sqrt{\varepsilon + m} w'^*(\sigma) \end{pmatrix} \quad (3.37)$$

Note that $w'^{(\sigma)} = -w'^{*}(\sigma)$, $\mathbf{n}_{-\mathbf{p}} = -\mathbf{n}_{\mathbf{p}}$, then

$$\begin{aligned}
U_T(\bar{u}_{-\mathbf{p}'\sigma})^T &= \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix} \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_{\mathbf{p}} \cdot \boldsymbol{\sigma}^*) w'^{(\sigma)} \\ \sqrt{\varepsilon + m} w'^{(\sigma)} \end{pmatrix} \\
&= \begin{pmatrix} \sqrt{\varepsilon - m} \sigma_y (\mathbf{n}_{\mathbf{p}} \cdot \boldsymbol{\sigma}^*) w'^{(\sigma)} \\ -\sqrt{\varepsilon + m} \sigma_y w'^{(\sigma)} \end{pmatrix} \\
&= \begin{pmatrix} \sqrt{\varepsilon - m} (\mathbf{n}_{\mathbf{p}} \cdot \boldsymbol{\sigma}) w^{(-\sigma)} \\ \sqrt{\varepsilon + m} w^{(-\sigma)} \end{pmatrix} \\
&= 2\sigma i u_{-p, -\sigma}
\end{aligned} \tag{3.38}$$

where we have used $w^{(\sigma)} = -2\sigma i w'^{(\sigma)}$. Similarly,

$$U_T(\bar{u}_{\mathbf{p}', -\sigma})^T = -2\sigma i u_{p\sigma} \tag{3.39}$$

Therefore,

$$\hat{\psi}^T = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(-2\sigma i \hat{a}_{-\mathbf{p}, -\sigma}^\dagger u_{p\sigma} e^{i\mathbf{p} \cdot \mathbf{r} - i\varepsilon t} + 2\sigma i \hat{b}_{-\mathbf{p}, -\sigma} u_{-p, -\sigma} e^{-i\mathbf{p} \cdot \mathbf{r} + i\varepsilon t} \right) \tag{3.40}$$

Comparing to the field operator before transformation,

$$\hat{\psi} = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\sigma} u_{p\sigma} e^{i\mathbf{p} \cdot \mathbf{r} - i\varepsilon t} + \hat{b}_{\mathbf{p}\sigma}^\dagger u_{-p, -\sigma} e^{-i\mathbf{p} \cdot \mathbf{r} + i\varepsilon t} \right) \tag{3.41}$$

we find the effect of time reversal on the field operator,

$$\boxed{T : \quad \hat{a}_{\mathbf{p}\sigma} \rightarrow -2\sigma i \hat{a}_{-\mathbf{p}, -\sigma}^\dagger, \quad \hat{b}_{\mathbf{p}\sigma} \rightarrow -2\sigma i \hat{b}_{-\mathbf{p}, -\sigma}^\dagger} \tag{3.42}$$

Time reversal interchanges the initial and final states, therefore the annihilation operators are replaced by creation operators. Time reversal also reverses both translational and rotational motions, thus changes the signs of both $\mathbf{p}$ and σ .

3.4 CPT Theorem

Now let's consider the consecutive operation of T , P , and C transformations. In spinor and standard representations,

$$\begin{aligned}
\hat{T}\psi(t, \mathbf{r}) &= i\gamma^3\gamma^1\gamma^0\bar{\psi}(-t, \mathbf{r}) = i\gamma^3\gamma^1\psi^*(-t, \mathbf{r}) \\
\hat{P}\hat{T}\psi(t, \mathbf{r}) &= i\gamma^0 (i\gamma^3\gamma^1\psi^*(-t, -\mathbf{r})) = -\gamma^0\gamma^3\gamma^1\psi^*(-t, -\mathbf{r}) \\
\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) &= \gamma^2 (-\gamma^0\gamma^3\gamma^1\psi^*(-t, -\mathbf{r}))^* = -\gamma^2\gamma^0\gamma^3\gamma^1\psi(-t, -\mathbf{r})
\end{aligned} \tag{3.43}$$

Using the commutation rules of the γ matrices, we find that

$$\boxed{\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) = i\gamma^5\psi(-t, -\mathbf{r})} \tag{3.44}$$

Further applying the commutation rule of γ^5 and γ^μ , it's easy to show that $\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r})$ satisfy the same Dirac equation as $\psi(t, \mathbf{r})$. We say that spin $\frac{1}{2}$ particles have CPT symmetry. In general, the **CPT theorem** states that, **the CPT symmetry holds for all relativistic invariant physical phenomena.**

In spinor representation, the transformations of 4-spinors ξ and η are,

$$\boxed{CPT : \quad \xi^\alpha \rightarrow -i\xi^\alpha, \quad \eta_{\dot{\alpha}} \rightarrow i\eta_{\dot{\alpha}}} \quad (3.45)$$

The transformation of the field operator in spinor or standard representation is given by

$$\boxed{\hat{\psi}^{CPT} = i\gamma^5 \hat{\psi}(-t, -\mathbf{r})} \quad (3.46)$$

Explicitly written,

$$\hat{\psi}^{CPT} = \sum_{\mathbf{p}, \sigma} \frac{1}{\sqrt{2\varepsilon}} \left(i\hat{a}_{\mathbf{p}\sigma} \gamma^5 u_{\mathbf{p}\sigma} e^{-i\mathbf{p}\cdot\mathbf{r} + i\varepsilon t} + i\hat{b}_{\mathbf{p}\sigma}^\dagger \gamma^5 u_{-\mathbf{p}, -\sigma} e^{i\mathbf{p}\cdot\mathbf{r} - i\varepsilon t} \right) \quad (3.47)$$

Following the same procedure used before, we calculate $\gamma^5 u_{\mathbf{p}\sigma}$ and $\gamma^5 u_{-\mathbf{p}, -\sigma}$ in the standard representation,

$$\gamma^5 u_{\mathbf{p}\sigma} = 2\sigma i u_{-\mathbf{p}\sigma}, \quad \gamma^5 u_{-\mathbf{p}, -\sigma} = 2\sigma i u_{\mathbf{p}, -\sigma} \quad (3.48)$$

Changing $\sigma \rightarrow -\sigma$ in the expression of $\hat{\psi}^{CPT}$ and comparing it with the expression of $\hat{\psi}$, we find the effect of CPT transformations on the field operator,

$$\boxed{CPT : \quad \hat{a}_{\mathbf{p}\sigma} \rightarrow 2\sigma \hat{b}_{\mathbf{p}, -\sigma}^\dagger, \quad \hat{b}_{\mathbf{p}\sigma} \rightarrow 2\sigma \hat{a}_{\mathbf{p}, -\sigma}^\dagger} \quad (3.49)$$

Note that the C , P , and T operations transform the coefficients ($\psi_{\mathbf{p}\sigma}$ and $\psi_{-\mathbf{p}, -\sigma}$) of the creation and annihilation operators, not the operator themselves. **These operator substitution rules only reflect the transformations of their coefficients.** Therefore, it is incorrect to consecutively apply the substitution rules of the creation and annihilation operators under T , P , and C transformations to get the substitution rule under the grouped CPT transformation. For example, using Eqs. (3.10), (3.24), and (3.42), we would arrive at

$$\hat{a}_{\mathbf{p}\sigma} \xrightarrow{T} -2\sigma i \hat{a}_{-\mathbf{p}, -\sigma}^\dagger \xrightarrow{P} -2\sigma \hat{a}_{\mathbf{p}, -\sigma}^\dagger \xrightarrow{C} -2\sigma \hat{b}_{\mathbf{p}, -\sigma}^\dagger \quad (3.50)$$

which differs from Eq. (3.49) by a sign. The problem is that, after the transformation T , $\hat{a}_{-\mathbf{p}, -\sigma}^\dagger$ is multiplied by $\psi_{\mathbf{p}\sigma}$ instead of $\bar{\psi}_{\mathbf{p}\sigma}$. Therefore, when P is applied, the effective substitution rule for $\hat{a}_{-\mathbf{p}, -\sigma}^\dagger$ is $\hat{a}_{-\mathbf{p}, -\sigma}^\dagger \rightarrow i\hat{a}_{\mathbf{p}, -\sigma}^\dagger$, instead of the complex conjugate of Eq. (3.10).

The grouped CPT transformation flips the particle's polarization but leaves the momentum unchanged. This is because the time reversal flips both the momentum and the polarization, then the flipped momentum is reverted by spatial inversion, whereas the polarization is unaffected by inversion. Table 1 summarized the **symmetry of position, momentum, polarization, and helicity under C , P , and T transformations.**

Table 1: Symmetry of Position, Momentum, Polarization, and Helicity under C , P , and T Transformations

	$\mathbf{r}$	$\mathbf{p}$	σ	λ
C	+1	+1	+1	+1
P	-1	-1	+1	-1
T	+1	-1	-1	+1
CPT	-1	+1	-1	-1